

Fairness-Aware VNF Mapping and Scheduling in Satellite Edge Networks for Mission-Critical Applications

Haftay Gebreslasie Abreha¹, Graduate Student Member, IEEE, Houcine Chougrani², Member, IEEE, Ilora Maity³, Member, IEEE, Youssouf Drif⁴, Member, IEEE, Christos Politis, Fellow, IEEE, and Symeon Chatzinotas⁵, Fellow, IEEE

Abstract—Satellite Edge Computing (SEC) is seen as a promising solution for deploying network functions in orbit to provide ubiquitous services with low latency and bandwidth. Software Defined Networks (SDN) and Network Function Virtualization (NFV) enable SEC to manage and deploy services more flexibly. In this paper, we study a dynamic and topology-aware VNF mapping and scheduling strategy within an SDN/NFV-enabled SEC infrastructure. Our focus is on meeting the stringent requirements of mission-critical (MC) applications, recognizing their significance in both satellite-to-satellite and edge-to-satellite communications while ensuring service delay margin fairness across various time-sensitive service requests. We formulate the VNF mapping and scheduling problem as an Integer Nonlinear Programming problem (INLP), with the objective of *minimax* fairness among specified requests while considering dynamic satellite network topology, traffic, and resource constraints. We then propose two algorithms for solving the INLP problem: Fairness-Aware Greedy Algorithm for Dynamic VNF Mapping and Scheduling (FAGD_MASC) and Fairness-Aware Simulated Annealing-Based Algorithm for Dynamic VNF Mapping and Scheduling (FASD_MASC) which are suitable for low and high service arrival rates, respectively. Our extensive simulations demonstrate that both FAGD_MASC and FASD_MASC approaches are very close to the optimization-based solution and outperform the benchmark solution in terms of service acceptance rates.

Index Terms—Satellite edge computing, software defined networking (SDN), network function virtualization (NFV), virtual network function (VNF), VNF scheduling, fairness.

I. INTRODUCTION

THE EVER-INCREASING demand for mission-critical (MC) applications in 6G has attracted the attention of

researchers [2]. MC applications typically deal with very serious situations, such as disaster management, rescue, and military operations, with stringent requirements such as latency and reliability, as their failure can result in the loss of lives and property [3]. Therefore, network architectures that provide high QoS and network coverage are crucial to meet the requirements. Terrestrial networks are vulnerable to natural disasters, such as earthquakes and hurricanes. In this context, satellite networks can be an alternative solution to overcome the challenges of network disruption.

MC applications depend on satellite networks during terrestrial communication failures, especially in scenarios requiring low-latency satellite-to-satellite links for real-time decision-making, such as emergency response operations [4]. The urgency of the services varies: for example, early warnings require rapid responses [5], while relief logistics can tolerate some delays [6]. Therefore, effective demand-driven service provisioning is crucial in satellite networks to accommodate these diverse requirements [7]. However, the traditional *bent-pipe* satellite architecture, which involves raw data transmission from satellites to remote terrestrial data centers without on-board processing, is not suitable for MC applications. This architecture suffers from limitations in latency and bandwidth consumption, hindering the timely delivery and processing of critical information, and impeding swift decision-making and response. Satellite Edge Computing (SEC) addresses this challenge by providing computing resources on-board in a *compute-as-a-service* model [8], [9], [10]. In this paradigm, satellites are equipped with microservers, enabling in-orbit data processing without relying on remote terrestrial data centers [8]. An SEC-enabled network facilitates ubiquitous, low latency, and bandwidth efficient service provisioning, making it particularly suitable for MC applications.

Despite the advantages of SEC in the delivery of services for MC applications, effectively managing the diverse and dynamic nature of network resources, such as computing, storage, and bandwidth, and traffic demands poses a challenge [10]. Network Function Virtualization (NFV) [11] and Software Defined Networking (SDN) [12] are two networking virtualization technologies that address this challenge by enabling agile and flexible network management and orchestrating service delivery. In an SDN/NFV-enabled environment, network services are represented as sequentially ordered

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Haftay Gebreslasie Abreha, Houcine Chougrani, Ilora Maity, Youssouf Drif, and Symeon Chatzinotas are with the Interdisciplinary Center for Security, Reliability and Trust, University of Luxembourg, 1855 Esch-sur-Alzette, Luxembourg (e-mail: haftay.abreha@uni.lu; houcine.chougrani@uni.lu; ilora.maity@uni.lu; youssouf.drif@uni.lu; symeon.chatzinotas@uni.lu).

Christos Politis is with the Department of Communications Systems, SES Techcom S.A., 6815 Betzdorf, Luxembourg (e-mail: christos.politis@ses.com).

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virtualized network functions (VNFs) known as Service Function Chaining (SFC) [13]. In an SEC-enabled network, an SFC request is completed by executing its VNF components on the SEC nodes on-board,¹ and selecting the actual path for SFC traffic to traverse between the source and destination nodes. The challenge of deploying VNF components for a service request in an SEC-enabled network with limited resources to meet the predefined requirements of MC applications is a significant research question. Addressing this challenge involves designing an appropriate *VNF mapping* and *scheduling* strategy [13]. The *VNF mapping* identifies optimal nodes for executing VNF components and the sets of physical links connecting these nodes, while *VNF scheduling* selects the most suitable time slots to process each VNF component on designated nodes.

Extensive research on the SEC [8], [9], [10], [14], [15], [16] has primarily focused on computational offloading strategies. However, optimal offloading performance relies on precise service provisioning, emphasizing the need for effective VNF mapping and scheduling strategies [13], [17], [18], [19], [20], [21], [22], [23]. Furthermore, except [13], the existing works emphasize VNF mapping, neglecting VNF scheduling, potentially leading to request rejections for violating service requirements, especially with multiple VNFs mapped to an SEC node. Current solutions use static approaches, causing inefficiencies and service blockage in dynamic traffic. Dynamic VNF mapping and scheduling improve resource use for better request acceptance, but in non-geostationary satellite (NGSO) systems, satellite mobility hampers system performance. Certain studies [1], [18], [21], [23] tackle the issue by discretizing the topology into quasi-static snapshots. They focus on link availability at a specific snapshot but overlook crucial details such as data rate and delay in link states while employing VNF mapping strategies. This approach may lead to performance degradation due to potential changes in link states between snapshots. To manage service requests spanning across multiple snapshots, some work [19] applies VNF re-mapping at the onset of each snapshot. However, in the resource-limited SEC environment, this method encounters challenges due to significant energy and computational costs [18], [24] associated with frequent reconfiguration.

Fairness is a crucial metric for fairly allocating resources like virtual machines, CPUs, and bandwidth in cloud computing systems [25], [26]. Although fairness can encompass various dimensions, our emphasis lies on ensuring fairness in terms of service delay margin. This choice is motivated by the unique demands of MC applications, which demand stringent latency requirements to ensure timely and effective responses to critical situations [27], [28]. Maintaining fairness of E2E service delays among requests can ensure service sensitivity and improve overall system QoS by treating time-critical and time-tolerant services differently. Furthermore, MC applications benefit from fairness by treating requests for services with the same delay requirements equally to ensure the QoS of

the system. Existing research on SEC and VNF mapping has often focused on computational offloading, network integration, and dynamic resource allocation. However, there is a gap in integrating MC requirements and fairness considerations into VNF mapping and scheduling in SEC networks.

In this context, the paper studies VNF mapping and scheduling strategies to maximize E2E delay margin² fairness among competing service requests. Through our work, we aim to enhance the reliability and efficiency of MC operations in an SEC-enabled network by considering both link availability and link state at each snapshot. Focusing on dynamic network architecture, our study facilitates interactions between ground-based users and satellites. This architecture integrates *satellite-to-satellite* collaborations and *edge-to-satellite* interactions for efficient service provisioning. Mathematically, we formulate a dynamic and topology-aware VNF mapping and scheduling as an integer nonlinear programming (INLP) problem to maximize the fairness of the delay margin among competing services, with specified *minimax* fairness. Initially, we address the problem with an optimization-based solution using the successive convex approximation (SCA) method, though it exhibits high time complexity. Thus, we propose two less complex algorithms to solve the problem: Fairness-Aware Greedy Algorithm for Dynamic VNF Mapping and Scheduling (FAGD_MASC) and Fairness-Aware Simulated Annealing-Based Algorithm for Dynamic VNF Mapping and Scheduling (FASD_MASC). Both algorithms perform similarly at lower traffic arrival rates, but FASD_MASC has a longer convergence time due to iterative and exhaustive searches in simulated annealing. Consequently, FAGD_MASC is more suitable for lower arrival rates. For fairness and acceptance rate, FASD_MASC outperforms FAGD_MASC at higher service arrival rates. Therefore, FASD_MASC is preferable in high arrival rate scenarios. The proposed solutions are evaluated through extensive simulations. Here is a summary of our main contributions.

- While existing works handle time-varying network topology by using sequential quasi-static topology snapshots, our approach goes further by conducting a comprehensive evaluation of link quality, considering both propagation delay and availability.
- We propose a novel integration of MC and fairness constraints into the dynamic VNF mapping and scheduling strategies within SEC networks. Our approach addresses the challenges posed by time-varying traffic workloads and changing network topologies, striving to achieve E2E service delay margin fairness. We first model the E2E delay and the safe delay margin taking into account the time-varying traffic requests and network topology challenges. We then introduce a dynamic VNF mapping and scheduling strategy in which the strategies are readjusted in response to traffic demand and network topology. Furthermore, we describe the network topology as a sequence of snapshots, where each snapshot shows a quasi-static topology. We then formulate a dynamic and

¹On-board or in-orbit computing is a future data center in orbit, where satellites act as small data centers and offer cloud-like services on board.

²An E2E delay margin of a service is defined as the safe margin to reach the upper bound of the delay required by the request.

topology-aware VNF mapping and scheduling strategy to maximize the delay margin fairness between competing services as an INLP problem with specified *minimax*.

- We solve the problem INLP using an optimization-based approach by applying the SCA method. Due to the time complexity of the optimization-based solution, we proposed two lower complex algorithms to solve the INLP problem: FAGD_MASC and FASD_MASC. We also evaluate the proposed algorithms compared to the optimization-based approach.
- We perform extensive simulations to evaluate the proposed solutions. The results show that the proposed strategies are close enough to the optimization-based solution and outperform the benchmark.

II. RELATED WORKS

A. Satellite Edge Computing

Research on computing services in space has become an increasingly hot topic in recent years [8], [9], [10], [14], [15], [16], [29]. To overcome the drawbacks of the bent pipe satellite network architecture, Denby and Lucia [9] proposed an orbital edge computing architecture that supports edge computing in nanosatellites for sensing, processing, and communication to maintain the required latency and energy. In [8], the *in-orbit computing as a service* model was presented to provide ubiquitous computing services to users. However, the above works [8], [9] focus on the placement of multi-access edge computing (MEC) servers on satellite nodes. Wang et al. [14] proposed a space edge computing architecture to deploy software in orbit and coordinate with cloud data centers. Furthermore, the authors of [10] discussed a satellite-terrestrial collaboration model to offload computations. However, none of them addressed the service delivery schemes, unlike our work.

Integrating network virtualization technologies like SDN/NFV with SEC-enabled networks enhances network management flexibility. In [15], the authors proposed an SDN/NFV-enabled virtualized resource and a multilayer architecture for integrated satellite-terrestrial edge computing networks. Yan et al. [16] introduced *5GsatEC*, an SEC framework, enhancing coverage and reducing service delay through embedded hardware and microservices on satellites. In [10] and [29], the authors discussed cooperative computing in a space-air-ground integrated network (SAGIN) architecture. Similarly, [3] studied integrating ultra-reliable, low-latency edge intelligence into the 6G cellular network for MC service support. The existing literature focused on computational offloading and integration with terrestrial networks. However, in the absence of an appropriate service provisioning scheme, offloading performance cannot be guaranteed. In contrast, our work emphasizes VNF mapping and scheduling strategies for service provisioning in an SEC-enabled network for MC applications.

B. VNF Mapping and Scheduling in SEC Environment

Wang et al. [17] demonstrated a reconfigurable SFC-based service provisioning framework in a SAGIN environment.

The authors defined the VNF mapping problem as INLP to maximize service acceptance rates while minimizing computational resource consumption. The work primarily explores VNF mapping in static scenarios. In this article, our focus is on dynamic VNF mapping, enabling the readjustment of the VNF mapping strategy for existing service requests in an SEC-enabled network. In [13], the authors introduced a dynamic VNF mapping and scheduling scheme within a SAGIN architecture for Internet of Vehicles (IoV) applications, formulated as a mixed-integer linear programming (MILP) problem. Unlike our work, they overlooked the impact of the time-varying satellite network topology, potentially causing variations in quality of service (QoS) that may not meet the stringent requirements for services in MC applications.

Several service provisioning schemes [18], [19], [20], [21], [22], [23], aimed to address challenges arising from the time-varying network topology inherent in NGSO constellations. Current methodologies often segment the NGSO topology into snapshots, with each snapshot portraying a quasi-static topology, primarily emphasizing link availability to distinguish between snapshots. However, knowing link availability alone is inadequate. Overlooking variations in link states (e.g., bandwidth, propagation delay) across snapshots may lead to E2E service delay requirement violations. Some literature [19], [20] addresses the impact of time-varying satellite network topology on system performance during VNF mapping, particularly for services spanning multiple snapshots. They proposed triggering a VNF migration and reconfiguration scheme in response to topology changes. In [19], the authors introduced a dynamic service migration and reconfiguration scheme based on satellite and ground user mobility for ensuring service continuity and meeting QoS requirements. Excluding [13], all existing work [18], [19], [20], [21], [22], [23] on service provisioning in satellite networks manage the VNF mapping strategy independently, neglecting the VNF scheduling strategy. In contrast, our work concentrates on topology-aware dynamic VNF mapping and scheduling schemes, discretizing the topology into snapshots while considering both link state and availability with prior knowledge of temporal variations.

Existing research in SEC and VNF mapping has often focused on computational offloading, network integration, and dynamic resource allocation. However, a gap exists in integrating MC requirements and fairness considerations into VNF mapping in SEC networks. While resource allocation is important, maintaining fairness in service delay margins for MC applications has been overlooked. Furthermore, existing solutions seldom address the time-varying topology of satellite networks, impacting the quality of service. Our work addresses these gaps by proposing dynamic, topology-aware optimizations for VNF mapping and scheduling. We aim to maximize fairness in delay margins by considering satellite network dynamics and MC requirements.

In summary, although there is some research on VNF mapping and scheduling in nonterrestrial networks (NTNs), more research is needed. Existing approaches in VNF mapping and scheduling often fall short in addressing the stringent requirements of MC applications. Our work aims to fill this gap

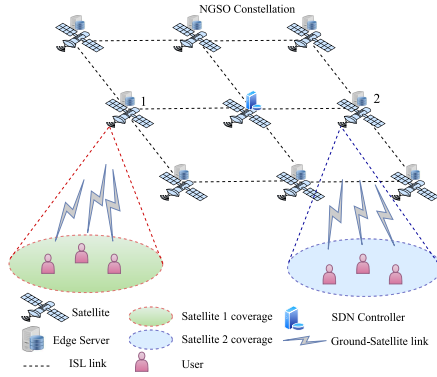


Fig. 1. SDN/NFV-enabled SEC network architecture.

by integrating service delay margin fairness with the specific demands of MC communication. Therefore, in this study, we focus mainly on the optimal selection of VNF mapping and scheduling strategies for dynamic traffic demands and topologies to meet the critical service requirements of MC applications in an MEC-enabled satellite network.

III. SYSTEM MODEL AND PROBLEM FORMULATION

This section presents the system model, encompassing both the substrate network and service requests. Figure 1 depicts the general SDN/NFV-enabled SEC network architecture for an NGSO satellite constellation, as elaborated in Section III-A. The NGSO architecture comprises MEC-enabled satellite nodes [8], [10], capable of hosting and processing VNFs to fulfill services with critical requirements in the orbit. This allows users to connect to SEC nodes via a wireless link and fulfill service requests in orbit. We assume that two adjacent SEC nodes are assumed to be connected by an inter-satellite link (ISL), like a wireless radio (RF) or free space optical (FSO) link. We consider an SDN controller to manage the VNF mapping and scheduling strategy, overseeing aspects such as VNF location, network topology, and resource availability. The controller is assumed to be located on one SEC-enabled satellite node in the constellation. However, the specific placement of an SDN controller for VNF mapping and scheduling in a satellite network requires further research and is beyond the scope of this study.

We assume a service request comprises the service type and source and destination addresses, detailed in Section III-B. The sources and destinations of service requests are users without access to the terrestrial network, communicating directly with satellites (e.g., satellites 1 and 2 in Figure 1) through wireless links. Tables I and II provide summaries of the relevant notation and decision variables used in this paper.

A. Substrate Network

We define the SEC-enabled network as an undirected graph $G = (V, E)$, where the set V contains physical nodes and E contains all possible links among the nodes. Additionally, V is subdivided into two subsets, namely, V_S and V_U , representing the SEC nodes and end users, respectively. Importantly, it

holds that $V = V_U \cup V_S$. Each SEC node $v_k \in V_S$ is characterized by its processing capacity (i.e., CPU) C_k and storage space (i.e., buffer size) β_k that can process VNFs and store the forwarding table for routing SFC traffic. End users act as both the source and destination devices for service requests, and it is assumed that they cannot process or store VNFs. The link $e_{h,k} \in E$ represents the connection between nodes $v_h \in V$ and $v_k \in V$. Each link $e_{h,k} \in E$ is characterized by its propagation delay $d_{h,k}$ and data rate capacity $b_{h,k}$. This study currently considers point-to-point (P2P) requests, with plans to explore multipoint-to-multipoint (MP-MP) slices in the future. The network architecture supports *satellite-to-satellite* and *edge-to-satellite* interactions, allowing efficient VNF processing. This facilitates efficient resource utilization by deploying various VNFs on different SEC nodes.

The satellite network topology in the NGSO constellation is time-varying due to satellite mobility. Consequently, link availability, delay, and rate vary, posing challenges for service provisioning. Fortunately, satellite mobility is predictable and periodic. To address this challenge, we subdivide the network topology into M sets of periodic quasi-static snapshots, represented as a set of time sequences of sub-topologies $G = \{G_1, \dots, G_t, \dots, G_M\}$, where G_t denotes the t^{th} snapshot. The number of snapshots within a recurrence period relies on the satellite mobility rate. Notably, these snapshots can be pre-calculated from the prior knowledge of the satellite ephemeris and user locations. Hence, we define the availability of the link $e_{h,k}$ at the snapshot time t , $\gamma_{h,k}(t)$, as:

$$\gamma_{h,k}(t) = \begin{cases} 1, & \text{if } e_{h,k} \in E \text{ is available at } t, \\ 0, & \text{Otherwise.} \end{cases} \quad (1)$$

Link availability in snapshots alone does not ensure QoS requirements due to variations in link states, impacting network performance and potentially causing incomplete or blocked service requests. To address this, we introduce a binary variable $\rho_{h,k}(t, t+1)$ to compare the propagation delay of the $e_{h,k}$ link between the snapshots G_t and the subsequent snapshot G_{t+1} , defined as:

$$\rho_{h,k}(t, t+1) = \begin{cases} 1, & \text{if } d_{h,k}(t) \geq d_{h,k}(t+1), \\ 0, & \text{Otherwise.} \end{cases} \quad (2)$$

where, $d_{h,k}(t)$ and $d_{h,k}(t+1)$ represent the propagation delay of link $e_{h,k}$ in snapshot G_t and subsequent snapshot G_{t+1} , respectively.

B. Service Request Model

We consider scenarios where terrestrial networks are unavailable, we rely exclusively on satellite networks as the primary infrastructure to provide MC services. This approach guarantees reliable connectivity and QoS communication for remote users, meeting the stringent requirements of MC services. We assume that service requests arrive randomly, each with unique requirements, service type, source, and destination address, following a Poisson distribution. Each service type comprises a sequentially ordered set of VNF components. Due to limited resources on SEC nodes, deploying all VNF types on a single node poses a significant

TABLE I
IMPORTANT PARAMETERS NOTATION AND DEFINITIONS

Notations	Definitions
V	Sets of SEC nodes
E	Sets of links between SEC nodes
v_k	k^{th} SEC nodes
$e_{h,k}$	Link between nodes, v_h and v_k
C_k	Processing capacity of node v_k
β_k	Memory space capacity of node v_k
G_t	t^{th} satellite network topology snapshot
$b_{h,k}(t)$	Data rate capacity of $e_{h,k}$ at G_t
$d_{h,k}(t)$	Propagation delay of $e_{h,k}$ at G_t
$\gamma_{h,k}(t)$	Availability of $e_{h,k}$ at G_t
$S(t)$	Sets of service requests at G_t
s_i	i^{th} service request
v_i^s	Source node where s_i starts
v_i^d	Destination node where s_i ends
F_i	Sets of VNF components of s_i
$f_{i,j}$	j^{th} VNF component of s_i
$\delta_k^{i,j}(t)$	$f_{i,j}$ support node v_k indicator
$\Gamma^i(t, t+1)$	s_i continuity to next snapshot indicator
$\rho_{h,k}(t, t+1)$	$e_{h,k}$ link state comparison between current and next snapshot
$A_{i,j}^t(t)$	Arrival of $f_{i,j}$ at v_k in snapshot, t
$\tau_{i,j}^k(t)$	queuing delay of $f_{i,j}$ at v_k in snapshot, t
$\epsilon_k^{i,j}(t)$	Processing delay of $f_{i,j}$ at node v_k
$d_i(t)$	E2E delay of service request, s_i
$\phi_i(t)$	Delay margin of s_i at snapshot, t
$\Phi^m(t)$	$\phi_i(t)$ component for VNF Mapping
$\Phi^s(t)$	$\phi_i(t)$ component for VNF scheduling

challenge. Consequently, different VNF components are distributed across various nodes based on resource availability. However, this distribution of VNF deployment introduces additional propagation delays, potentially violating the service requirements. To address this challenge, careful VNF mapping is required.

We consider $S(t)$ as a set of active MC service requests in the system (*i.e.*, newly arrived or currently in-process) at the snapshot t . As mentioned earlier, we assume that users generate and receive SFC request traffic. A service request comprises the source and destination addresses, representing the locations where the service originates and finishes. Additionally, it includes the service type, which defines the nature of the service request and its requirements. Each service request $s_i \in S(t)$, is represented as $s_i = \langle v_i^s, Q_i, v_i^d \rangle$, where Q_i denotes the service type, and $v_i^s, v_i^d \in V_u$ are the source and destination user addresses, respectively. The service type Q_i is expressed as $Q_i = \langle F_i, C_i, \beta_i, B_i, D_i \rangle$. Here, F_i is the ordered set of VNF components of Q_i , with $f_{i,j} \in F_i$ representing the j^{th} VNF component in Q_i . The notation $c_{i,j}$ and $\beta_{i,j}$ denote the *computing* and *storage space* resources required to execute $f_{i,j}$. B_i and D_i are the minimum E2E bandwidth and maximum E2E delay tolerated by the service request s_i , respectively. These parameters ensure that MC service requests exhibit diverse requirements in terms of bandwidth and delay, with specific services demanding high E2E bandwidth for rapid data transmission and others requiring stringent delay constraints for real-time communication. All relevant notations are listed in Tables I and II.

The decision of mapping the VNF $f_{i,j}$ to the physical node v_k at snapshot t is represented by a binary variable, defined as:

$$X_k^{i,j}(t) = \begin{cases} 1, & f_{i,j} \text{ is mapped onto node } v_k \text{ at } t, \\ 0, & \text{Otherwise.} \end{cases} \quad (3)$$

Furthermore, we assume each SEC node can support specific VNF types due to the diverse resource requirements and

TABLE II
DECISION VARIABLES

Notations	Definitions
$X_k^{i,j}(t)$	Binary variable indicating if the VNF $f_{i,j}$ is mapped on the SEC node v_k at G_t .
$Y_{(h,k)}^{i,j}(t)$	Binary variable indicating if the virtual link between the VNFs $f_{i,j-1}$ and $f_{i,j}$ mapping includes physical link between the nodes of v_h and v_k
$Z_{ij,gq}^k(t)$	Binary variable indicating if the VNF $f_{i,j}$ has to wait compared to the corresponding VNF $f_{g,q}$ to complete processing at node v_k at G_t

operational characteristics of the VNFs. We describe a binary variable $\delta_k^{i,j}(t) \in \{0, 1\}$ to describe the VNF supporting node indicator as follows:

$$\delta_k^{i,j}(t) = \begin{cases} 1, & \text{if } f_{i,j} \text{ is supported by node } v_k \text{ at } t, \\ 0, & \text{Otherwise} \end{cases} \quad (4)$$

We consider that a virtual link between consecutive VNF components of a service request can be mapped onto a path with multiple physical links in the absence of direct connections between the nodes hosting the VNFs. Therefore, we define the decision to map the virtual link connecting VNFs $f_{i,j} \in F_i$ and $f_{i,j-1} \in F_i$ to the physical path that comprises link $e_{h,k} \in E$ in G_t as:

$$Y_{h,k}^{i,j}(t) = \begin{cases} 1, & \text{If virtual link between VNFs } f_{i,j} \text{ and } f_{i,j-1} \text{ is mapped to a physical path that includes the link } e_{h,k} \text{ at snapshot } t, \\ 0, & \text{Otherwise.} \end{cases} \quad (5)$$

We assume the sets of physical links $E_{h,k}^{sh} \in E$, which form the shortest physical path connecting the nodes $v_h \in V$ and $v_k \in V$. This path can be computed using a shortest path algorithm, such as the Dijkstra algorithm.

An efficient VNF scheduling scheme is necessary when the resources on a node are insufficient to process concurrent service requests. The decision of processing order for concurrent VNF components of competing service requests $s_i \in S(t)$ and $s_g \in S(t)$, represented by $f_{i,j} \in F_i$ and $f_{g,q} \in F_g$ respectively, at node v_k and snapshot t is defined as:

$$Z_{ij,gq}^k(t) = \begin{cases} 1, & \text{if } f_{i,j} \text{ processes at } v_k \text{ after } f_{g,q} \text{ at } t, \\ 0, & \text{Otherwise.} \end{cases} \quad (6)$$

Service requests may start and end at a specific snapshot, t , or extend to the next snapshot, $t+1$. The service continuity indicator, $\Gamma^i(t, t+1)$, indicates if service request s_i spans from the t^{th} snapshot to the subsequent $(t+1)^{th}$ snapshot and is defined as:

$$\Gamma^i(t, t+1) = \begin{cases} 1, & \text{if } s_i \text{ starts at } t \text{ and continues in } t+1, \\ 0, & \text{Otherwise.} \end{cases} \quad (7)$$

C. E2E Service Delay and Service Delay Margin Model

In our work, we consider the E2E service delay to be composed of three fundamental components: 1) propagation delay, 2) processing delay, and 3) queuing delay. The propagation delay accounts for the cumulative propagation delay of the selected links to fulfill a given request. Similarly, the processing delay is the total delay required to execute the VNF components of the service request. Additionally, the queuing delay represents the overall delay for the VNF components of

a given service request to wait before starting execution due to resource limitations for concurrent VNF processing. Please refer to Appendix A for a detailed explanation of delay components. Notably, our analysis excludes the deployment time required for the SDN/NFV orchestrator, as it falls beyond the scope of this study.

Within this context, the service delay comprises two primary components: the VNF mapping delay, encompassing processing and propagation delays, and the VNF scheduling delay, involving queuing delay. The VNF scheduling delay for a service request is determined by its processing delay and the arrival time of competing VNFs at the selected nodes for processing its VNF components. Thus, $A_{ij}^k(t)$, denoting the arrival time of VNF $f_{i,j}$ at node v_k , is considered when its successive VNF $f_{i,j-1}$ is mapped to node v_h at snapshot t (i.e., $X_h^{i,j-1}(t) = 1$). This is defined as:

$$A_{i,j}^k(t) = \left(A_{i,j-1}^h(t) + \ell_h^{i,j-1} \right) X_h^{i,j-1}(t) + d_{h,k}(t), \quad (8)$$

where $A_{i,j-1}^h(t)$ and $\ell_h^{i,j-1}$ represent the arrival time and processing delay of the preceding VNF component, $f_{i,j-1}$, in the service s_i at its designated node, v_h , respectively.

We accordingly model the E2E service delay of a service request s_i at a given snapshot t $d_i(t)$ as the sum of the VNF mapping delay $d_i^m(t)$ and VNF scheduling delay $d_i^s(t)$.

$$d_i(t) = d_i^m(t) + d_i^s(t), \quad (9)$$

See Appendix A for details of $d_i(t)$, $d_i^m(t)$, and $d_i^s(t)$.

Remark 1: Propagation delay usually dominates the service delay, but when multiple VNFs are assigned to limited nodes, the queueing delay can become comparable.

Definition 1: A service request s_i delay margin is the safe margin between the maximum tolerable delay D_i and the actual E2E service delay $d_i(t)$ at a specific snapshot, t :

$$\phi_i(t) = \frac{d_i(t)}{D_i}, \quad (10)$$

Using Equation (9), we redefine (10) as the sum of delay margins related to VNF mapping and scheduling as:

$$\Phi_i(t) = \Phi_i^m(t) + \Phi_i^s(t), \quad (11)$$

where $\Phi_i^m(t)$ and $\Phi_i^s(t)$ are the delay margins related to VNF mapping and scheduling, calculated as $\Phi_i^m(t) = \frac{d_i^m(t)}{D_i}$ and $\Phi_i^s(t) = \frac{d_i^s(t)}{D_i}$. The set of delay margins for competing services is denoted as $\Phi(t) = \{\phi_1(t), \dots, \phi_i(t), \dots, \phi_{|S(t)|}(t)\}$. Using Equation (11), it can be further written as:

$$\Phi(t) = \Phi^m(t) + \Phi^s(t), \quad (12)$$

where $\Phi^m(t)$ and $\Phi^s(t)$ are sets of delay margins for VNF mapping and scheduling, respectively.

D. Problem Formulation

We formulate the optimization problem to maximize fairness in the delay margin among competing MC service requests. Decision variables $\mathbf{X}(t)$, $\mathbf{Y}(t)$, and $\mathbf{Z}(t)$, representing VNF node mapping, virtual link mapping, and VNF scheduling in snapshot t , are involved. The goal is to minimize $\phi_i(t)$,

the delay margin for the most adversely treated service request, s_i (i.e., minimize the maximum), among competing services. This improves overall system performance and ensures a balanced distribution of delay margins. The objective function can be defined as *Min-Max fairness* of the delay margin, $\Phi(t)$ at snapshot t .

$$\text{minimize } \{ \max_i \Phi(t) \} \quad (13a)$$

$$\text{s. t. } X_k^{i,j}(t) \in \{0, 1\}, \forall f_{i,j} \in s_i, \forall v_k \in V, \quad (13b)$$

$$Y_{h,k}^{i,j}(t) \in \{0, 1\}, \forall f_{i,j} \in s_i, \forall e_{h,k} \in E, \quad (13c)$$

$$Z_{ij,gq}^k \in \{0, 1\}, \forall f_{i,j} \in s_i, f_{g,q} \in s_g, \forall v_k \in V, \quad (13d)$$

$$\sum_{v_k \in V} X_k^{i,j}(t) = 1, \forall f_{i,j} \in s_i, \quad (13e)$$

$$X_k^{i,j}(t) \leq \delta_k^{i,j}(t), \forall f_{i,j} \in s_i, \quad (13f)$$

$$|S(t)| |F_i| \sum_{i=1} \sum_{j=1} X_k^{i,j}(t) c_{i,j} \leq C_k, \forall v_k \in V, \quad (13g)$$

$$|S(t)| |F_i| \sum_{i=1} \sum_{j=1} X_k^{i,j}(t) \beta_{i,j} \leq \beta_k, \forall v_k \in V, \quad (13h)$$

$$\sum_{e_{h,k} \in E} Y_{h,k}^{i,j}(t) \gamma^{h,k}(t) \geq 1, \forall f_{i,j} \in s_i, \quad (13i)$$

$$|S(t)| |F_i| \sum_{i=1} \sum_{j=1} Y_{h,k}^{i,j}(t) B_i \gamma^{h,k}(t) \leq b_{h,k}(t), \forall e_{h,k} \in E, \quad (13j)$$

$$A_{i,j}^k(t) X_k^{i,j}(t) \geq A_{i,j}^h(t) X_h^{i,j-1}(t), \forall X_h^{i,j-1}(t) = 1, \quad (13k)$$

$$Y_{h,k}^{i,j}(t) \leq \gamma_{h,k}(t) (1 - \Gamma^i(t, t+1) + \gamma_{h,k}(t+1) \rho_{h,k}(t, t+1)), \quad (13l)$$

$$Z_{ij,gq}^k(t) + Z_{gq,ij}^k(t) = 1, \forall \{g \neq i \text{ OR } q \neq j, v_k \in V\}, \quad (13m)$$

$$Y_{m,n}^{i,j}(t) \geq X_h^{i,j-1}(t) + X_k^{i,j}(t) - 1, \forall e_{m,n} \in E_{h,k}^{sh}, \quad (13n)$$

$$Y_{m,n}^{i,j}(t) \leq \sum_{v_h, v_k \in V} (X_h^{i,j-1}(t) + X_k^{i,j}(t)) - 1, \quad (13o)$$

$$Y_{m,n}^{i,j}(t) = 0, \forall e_{m,n} \notin E_{h,k}^{sh}, X_h^{i,j-1}(t) = 1, X_k^{i,j}(t) = 1, \quad (13p)$$

Constraints (13b), (13c), and (13d) indicate binary decision variables. Constraint (13e) ensures that a VNF component of a service request is processed by only one node. Constraint (13f) specifies that a VNF can only be mapped to a node that has the ability to execute it. Constraints (13g) and (13h) assert that a VNF can only be mapped to a node with sufficient storage and processing capacity, respectively. Constraint (13i) conveys that every virtual link must be mapped to at least one physical link that exists in a given snapshot. Constraint (13j) maintains that a virtual link can only be mapped to links with sufficient bandwidth for the VNF to transmit. Constraint (13k) requires the sequential processing of VNF components within a service request, ensuring adherence to their specified order for completion. Constraint (13l) allows assigning a service from snapshot G_t to G_{t+1} to a link if two conditions are met: the link must exist in both snapshots, and its propagation

delay in G_{t+1} should not exceed that in G_t . Constraint (13m) guarantees that a node processes only one VNF at a time. In case of two simultaneous requests, it dictates processing one while the other awaits execution, maintaining a specific order. Constraints (13n)-(13p) ensure that the virtual link between consecutive VNF components in a service request is assigned to all physical links along the shortest path connecting their hosting nodes. The Constraint (13o) affirms the precise mapping of the virtual link, accommodating scenarios in which a physical link is part of multiple shortest paths or a single shortest path between SEC nodes. Constraint (13p) ensures that a virtual link cannot be mapped to a physical link that is not part of the shortest path between the nodes hosting two consecutive VNF components of a service request.

The optimization problem formulated in Equation (13) falls within the class of INLP problems. The nonlinearity arises from the product of decision variables $\mathbf{X}(t)$ and $\mathbf{Y}(t)$ in Equation (24) from Appendix A. Furthermore, the node mapping, $X_k^{i,j}$, in Equations (8) and (24), depends on the preceding VNF mapping $X_h^{i,j-1}$. This complexity makes the optimization problem computationally expensive to solve. Consequently, we decompose the problem into two stages: the VNF mapping optimization problem (P1) and the VNF scheduling optimization problem (P2). This decomposition linearizes the problem, allowing for the analysis of problem (13) as two linear subproblems, P1 and P2, described in Equations (14) and (15), respectively. P1 represents the VNF mapping optimization problem with decision variables $\mathbf{X}(t)$ and $\mathbf{Y}(t)$ as defined below.

$$\begin{aligned} \text{P1 : } & \text{minimize } \left\{ \max_i \Phi^m(t) \right\} \\ & \text{s. t., } (13b), (13c), (13e) - (13l), (13n) - (13p), (14) \end{aligned}$$

Furthermore, P2 is the VNF scheduling optimization problem with the decision variable $\mathbf{Z}(t)$ as defined below.

$$\begin{aligned} \text{P2 : } & \text{minimize } \left\{ \max_i \Phi^s(t) \right\} \\ & \text{s. t., } (13d), (13m), (15) \end{aligned}$$

We address the challenge of the sequential dependence of the variable $\mathbf{X}(t)$ initially by solving P1 (i.e., $\mathbf{X}^*(t), \mathbf{Y}^*(t)$) and then solving P2 (i.e., $\mathbf{Z}^*(t)$). In this case, the solution to problem P1 is the input to problem P2. However, due to the binary variables of $\mathbf{X}(t)$, $\mathbf{Y}(t)$, and $\mathbf{Z}(t)$; P1 and P2 are combinatorial problems. Inspired by [30], we apply a binary relaxation technique to solve the subproblems efficiently where the binary variables are relaxed to satisfy $0 \leq \mathbf{X}(t) \leq 1$, $0 \leq \mathbf{Y}(t) \leq 1$, and $0 \leq \mathbf{Z}(t) \leq 1$. Moreover, it is essential to employ an efficient and suitable relaxation method to ensure that the relaxed formulations of P1 and P2 produce output variables $\mathbf{X}^*(t), \mathbf{Y}^*(t), \mathbf{Z}^*(t)$ constrained to binary values of either 1 or 0, as stated in Equations (17) and (17). We transform the problems P1 and P2 into the convex-concave procedure (CCP) P3 and P4, respectively [31] by carefully selecting the penalty function and parameters so that can be solved directly using mathematical optimization solvers such as CVX [32] iteratively.

$$\begin{aligned} \text{P3 } & \text{minimize } \left\{ \max_i [\Phi^m(t) - (-\psi_{xy})] \right\} \\ & \text{s. t., } 0 \leq \mathbf{X}(t) \leq 1, 0 \leq \mathbf{Y}(t) \leq 1, \\ & \quad (13e) - (13l), (13n) - (13p), (16) \end{aligned}$$

$$\begin{aligned} \text{P4 } & \text{minimize } \left\{ \max_i [\Phi^s(t) - (-\psi_z)] \right\} \\ & \text{s. t., } 0 \leq \mathbf{Z}(t) \leq 1, (13m), (17) \end{aligned}$$

Here, ψ_{xy} and ψ_z are the penalty functions of the original VNF mapping and scheduling optimization problems, respectively. The details of the penalty functions and parameters for binary relaxation can be seen in the appendix B. The subproblems P3 and P4 are convex problems that can be effectively solved using standard optimization software tools. Furthermore, the problems P3 and P4 are classified as difference of convex (DC) programming problems. Fortunately, these problems can be effectively solved using an iterative algorithm based on the convex-concave procedure (CCP) [31]. The CCP algorithm provides a systematic approach to iteratively handle the convex and concave components of the objective function, leading to convergence to optimal solutions for the DC programming problems. Therefore, we apply the decomposition and relaxation methods to problem (13) and an optimization-based approach based on SCA to solve them iteratively. We solve P3 to obtain the optimal solution $\mathbf{X}^*(t)$ and $\mathbf{Y}^*(t)$ and update X^α and Y^α by using CVX, where X^α and Y^α are the VNF node mapping and virtual link mapping solutions in the α^{th} iteration, respectively. Then we solve P4 for a given X^α and Y^α to obtain the optimal solution $\mathbf{Z}^*(t)$ using CVX and update Z^α by $\mathbf{Z}^*(t)$, where Z^α is the VNF scheduling solution in the α^{th} iteration. This process repeats until the solution converges by updating $\alpha = \alpha + 1$ and finally, we obtain the optimal solution $\mathbf{X}^*(t), \mathbf{Y}^*(t),$ and $\mathbf{Z}^*(t)$. However, the high complexity of the optimization-based solution makes it unsuitable for MC applications. Typically, VNF mapping and scheduling problems are NP-hard, inspiring the common use of heuristic algorithms for resolution [21], [22]. Consequently, we present two less complex algorithms suited for different traffic arrival rates to address the INLP problem.

IV. FAIRNESS-AWARE DYNAMIC VNF MAPPING AND SCHEDULING FOR TIME-VARYING SATELLITE NETWORK

In this section, we introduce two heuristic algorithms, namely FAGD_MASC and FASD_MASC, to efficiently obtain near-optimal solutions for the INLP problem in low and high traffic demand scenarios, respectively. Initially, the algorithms configure VNF mapping and then initiate a VNF scheduling strategy based on delay margin values. They dynamically adapt to changing traffic demands through VNF re-mapping and re-scheduling, triggered by new service requests and E2E service delay violations. Further details are provided in the next subsections.

A. Feasible VNF Mapping Strategy Generator Algorithm

Algorithm 1 initializes the VNF mapping strategy $\pi_{x,y} = \{\mathbf{X}(t), \mathbf{Y}(t)\}$ for the proposed solutions FAGD_MASC and FASD_MASC as described in Section IV-B and C, respectively. Here, $\mathbf{X}(t)$ and $\mathbf{Y}(t)$ are the set of nodes and virtual

Algorithm 1 FEASIBLE VNF MAPPING STRATEGY GENERATOR ALGORITHM

Input: SFC request s_i in snapshot G_t
Output: $\pi_{x,y} \leftarrow \{X(t), Y(t)\}$ $\triangleright$ VNF Mapping strategy
 1: Extract the VNF components F_i of the service request s_i ,
 2: **for** each $f_{i,j} \in F_i$ **do**
 3: $\mathcal{N}(f_{i,j}) \leftarrow$ Sets of SEC nodes that can process the VNF, $f_{i,j}$
 4: Select a node $v_k \in \mathcal{N}(f_{i,j})$ randomly satisfying Constraints (13g) and (13h)
 5: $X_k^{i,j}(t) \leftarrow 1$ $\triangleright$ Update the VNF node mapping strategy
 6: **if** $X_k^{i,j-1}(t) == 1$ **and** $v_h \in \mathcal{N}(f_{i,j-1})$ **then**
 7: Find the shortest path between v_h and v_k satisfying Constraints (13i), (13j), (13l), (13n)-(13p) using Dijkstra's algorithm.
 8: $Y_{h,k}^{i,j}(t) \leftarrow 1$ $\triangleright$ Update the Virtual link mapping strategy
 9: $\pi_{x,y} \leftarrow \{X_k^{i,j}(t), Y_{h,k}^{i,j}(t)\}$ $\triangleright$ Update the VNF mapping for $f_{i,j}$
 10: **end if**
 11: **end for**
 12: **return** $\pi_{x,y}$

link mapping strategies, respectively, of competing services in the system $S(t)$. Given a service request $s_i \in S(t)$ as an input parameter in the current snapshot t , the algorithm returns the VNF mapping strategy $\pi_{x,y}$.

First, the algorithm extracts the VNF component F_i of the service request s_i (line 1). The next step is to identify and group the specific nodes that support the VNF. For example, $\mathcal{N}(f_{i,j})$ denotes the sets of all nodes that can support and process the VNF $f_{i,j}$ (line 3). For each VNF component of the service request s_i , an SEC node is randomly selected from the corresponding VNF supporting nodes that satisfy the Constraints (13f), (13g) and (13h), (line 4). The VNF mapping strategy is updated for each VNF component in the service request. For example, for VNF $f_{i,j} \in F_i$ on an SEC node $v_k \in \mathcal{N}(f_{i,j})$ in G_t , it is denoted as $X_k^{i,j}(t) = 1$ (line 5). The algorithm selects the shortest path between nodes hosting consecutive VNF components in the service request, adhering to Constraints (13i) and (13j). Moreover, to account for services continuing from the current snapshot t to the next snapshot $t+1$ (i.e., the snapshot in which the VNF mapping decision is made), the selected links must satisfy the constraint specified in Equation (13l). Therefore, the Dijkstra algorithm is used to determine the shortest path, adhering to Constraints (13n)-(13o) (line 7). The algorithm updates the corresponding node and link of the VNF mapping strategies $X(t)$ and $Y(t)$ (lines 8 and 9) and returns the updated $\pi_{x,y}$ (line 12).

B. Fairness-Aware Greedy Algorithm for Dynamic VNF Mapping and Scheduling (FAGD_MASC)

In this subsection, we propose FAGD_MASC to solve the INLP problem, as outlined in Algorithm 2. The algorithm takes the set of competing MC service requests $S(t)$ in snapshot t as input and outputs a sub-optimal VNF mapping and scheduling strategy $\pi^*(t)$. First, the VNF mapping strategy $\pi_{x,y}^0(t)$ is initialized using Algorithm 1 for the sets of service requests $S(t)$ (line 1). Then, the corresponding VNF scheduling in the selected VNF mapping and scheduling strategy, $\pi_z^0(t)$, is calculated (line 2). Furthermore, the optimal VNF mapping and scheduling strategy $\pi^*(t)$ is updated by the initial strategy $\{\pi_{x,y}^0(t), \pi_z^0(t)\}$ (line 3). Similarly, the maximum delay margin $\phi^*(t)$ is also updated from the optimum strategy obtained (line 4).

Algorithm 2 PROPOSED FAGD_MASC ALGORITHM

Input: Sets of competing MC service requests, $S(t)$
Output: $\pi^*(t) \leftarrow \{X^*(t), Y^*(t), Z^*(t)\}$ $\triangleright$ Optimal VNF Mapping and Scheduling
 1: $\pi_{x,y}^0(t) \triangleright$ Initialize the VNF mapping strategy using Algorithm 1
 2: $\pi_z^0(t) \triangleright$ Compute the corresponding VNF scheduling on $\pi_{x,y}^0(t)$
 3: $\pi^*(t) \leftarrow \{\pi_{x,y}^0(t), \pi_z^0(t)\}$ $\triangleright$ Update the optimal VNF mapping and scheduling
 4: $\phi^*(t) \leftarrow \max_i \{\Phi(t)\}$
 5: **while** StopConditionNotMet() **do**
 6: **for** each competing service request $s_i \in S(t)$ **do**
 7: **if** $E_i^p(t) == \emptyset$ **then**
 8: Trigger VNF Re-mapping and Re-scheduling function
 9: **end if**
 10: $\pi_{x,y}(t) \triangleright$ Generate a VNF mapping using Algorithm 1
 11: $\pi_z(t) \leftarrow Z(t) \triangleright$ Compute the corresponding VNF scheduling from $\pi_{x,y}(t)$
 12: $\pi(t) \leftarrow \{\pi_{x,y}(t), \pi_z(t)\}$
 13: $\phi(t) \leftarrow \max_i \{\Phi(t)\}$
 14: **if** $\phi(t) \leq \phi^*(t)$ **then**
 15: $\pi^*(t) \leftarrow \pi(t)$
 16: $\phi^*(t) \leftarrow \phi(t) \triangleright$ Update the optimal delay margin
 17: **end if**
 18: **end for**
 19: **end while**
 20: Update the available resources of the network
 21: **return** $\pi^*(t)$

For each service request $s_i \in S(t)$, the algorithm first checks if there exists a path $E_i^p(t) \in E$ with a set of physical links that meet the Constraints (13i), (13j), and (13l) (line 7). When there is no possible path to accommodate the service request s_i (i.e., $E_i^p(t) = \emptyset$), the algorithm invokes the VNF re-mapping and re-scheduling strategy (line 8). Otherwise, if there are feasible paths ($E_i^p(t) \neq \emptyset$), the algorithm calculates the VNF mapping for the service request s_i using Algorithm 1 (line 10). The algorithm FAGD_MASC computes the VNF scheduling for each VNF component F_i based on the service delay margin $\Phi(t)$ (line 11). The algorithm then updates the VNF mapping and scheduling $\pi(t)$ (line 12). In addition, it updates the maximum service delay margin among service requests ($\phi(t) = \max_i \{\Phi(t)\}$) (line 13). FAGD_MASC updates the optimal VNF mapping and scheduling, $\pi^*(t)$ with the current, $\pi(t)$ if and only if the current maximum delay margin is less than or equal to that of the previous iteration (i.e., $\phi(t) \leq \phi^*(t)$), where $\phi^*(t)$ is the maximum optimal delay margin obtained in the previous (lines 14-17).

The process continues until the stop condition is reached, determined by the StopConditionNotMet() function (line 5). The iteration stops if: 1) the maximum number of iterations is reached, 2) the objective function value ($\phi^*(t)$) remains unchanged, or 3) no candidate path in $E_i^p(t)$ satisfies the constraints after VNF re-mapping and re-scheduling (lines 5-19). Finally, the algorithm returns the optimal VNF mapping and scheduling strategy $\pi^*(t)$ at snapshot time, t , updating the network resources (lines 20-21).

C. Fairness-Aware Simulated Annealing-Based Algorithm for Dynamic VNF Mapping and Scheduling (FASD_MASC)

FASD_MASC algorithm is a Simulated Annealing (SA)-based approach [33] proposed to solve the INLP problem. SA is inspired by metallurgical annealing, where controlled heating and cooling create a crystalline structure with minimal

Algorithm 3 PROPOSED FASD_MASC ALGORITHM

Input: $S(t), \pi^0(t), T_0, r, L, p$
Output: $\pi(t)$ $\triangleright$ Final VNF Mapping and Scheduling strategy

- 1: $T \leftarrow T_0$ $\triangleright$ Initialize current temperature
- 2: $\pi \leftarrow \pi^0(t)$ $\triangleright$ Initiate the current VNF mapping strategy state
- 3: **while** $T > 0$ **do**
- 4: **while** $L > 0$ **do**
- 5: $\pi^{Next} \leftarrow \text{GenerateNextStrategy}(\pi)$ $\triangleright$ Next VNF mapping and scheduling
- 6: **if** $\exp(\frac{\text{fitness}(\pi) - \text{fitness}(\pi^{Next})}{\mathcal{B}T}) > p$ **then**
- 7: $\pi \leftarrow \pi^{Next}$ $\triangleright$ Update the current VNF mapping and scheduling
- 8: **end if**
- 9: $L \leftarrow L - 1$
- 10: **end while**
- 11: $T \leftarrow r * T$
- 12: **end while**
- 13: Update the available resources of the network
- 14: **return** $\pi^*(t) \leftarrow \pi$

errors. Our choice of SA is driven by its simplicity and maturity as a metaheuristic algorithm, with theoretical convergence proofs [33].

FASD_MASC algorithm generates sub-optimal VNF mapping and scheduling decisions based on SFC requests, available network resources, and network topology over the snapshots. Given the set of MC service requests $S(t)$ and the initial state of the VNF mapping and scheduling strategy state $\pi^0(t)$ in snapshot G_t , it returns the optimum VNF mapping and scheduling strategy $\pi^*(t)$ as the final state, considering the current state. Algorithm 3 outlines the steps of the proposed FASD_MASC algorithm. Furthermore, the SA parameters, including the initial temperature T_0 , cooling rate r , Markov chain length L , and acceptance probability p , are also input parameters for the FASD_MASC algorithm. These parameters significantly impact the algorithm's performance, necessitating careful selection [33]. T_0 determines the speed of convergence to the global optimum: higher values lead to slower convergence, while lower values risk premature convergence to a local optimum. The cooling rate r determines the rate of temperature reduction at each iteration until T reaches zero, indicating the end of the iterations. Similarly, the Markov chain length L specifies the number of iterations before a temperature decreases. Furthermore, the acceptance probability p determines the acceptability of new solutions.

The FASD_MASC algorithm starts by initializing the current temperature T to the initial temperature T_0 (line 1). It uses Algorithm 1 to create the initial VNF mapping strategy. A demand-driven approach based on the delay margin $\phi(t)$ generates a VNF scheduling strategy for active service requests $S(t)$, updating the initial VNF mapping and scheduling strategy $\pi^0(t)$ at snapshot t . This strategy serves as the initial state of the process. Then, the current VNF mapping and scheduling strategy π is set as the current state, initialized using the initial state $\pi^0(t)$ (line 2). The main goal of the FASD_MASC algorithm is to determine an optimal VNF mapping and scheduling strategy that improves delay margin fairness among competing MC service requests. The algorithm iterates a maximum of L times for each value of current temperature T . Within each iteration, the algorithm generates the next state π^{Next} using Algorithm 4 based on the current state

Algorithm 4 GENERATENEXTSTRATEGY

Input: π $\triangleright$ Current VNF Mapping and Scheduling strategy
Output: π^{Next} $\triangleright$ Next VNF Mapping and Scheduling strategy

- 1: Randomly select a service request s_i from the set of active MC service requests $S(t)$ in the current state, π .
- 2: Randomly choose a VNF component $f_{i,j}$ associated with s_i .
- 3: Identify the source node v_k to which $f_{i,j}$ is currently mapped ($x_k^{i,j} = 1$) in the current state, π .
- 4: Determine potential target nodes $\mathcal{N}(f_{i,j})$ for relocation, considering resource availability and constraints.
- 5: Randomly select a suitable target node v_h from $\mathcal{N}(f_{i,j})$.
- 6: Perform the relocation operation: Update the VNF mapping, X^{Next} , from the current state, π , by setting $x_k^{i,j}$ to 0 and $x_h^{i,j}$ to 1.
- 7: Adjust virtual link mapping, Y^{Next} from X^{Next} to account for the relocation.
- 8: Adjust VNF scheduling, Z^{Next} from X^{Next} and Y^{Next}
- 9: **return** $\pi^{Next} \leftarrow \{X^{Next}, Y^{Next}, Z^{Next}\}$

Algorithm 5 FITNESS COMPUTATION

Input: π $\triangleright$ VNF Mapping and Scheduling strategy
Output: $\text{fitness}(\pi)$ $\triangleright$ fitness for π

- 1: Compute the delay margin Φ from π for requests using Equation (12).
- 2: $\phi_{max} \leftarrow \max_i \{\Phi\}$ $\triangleright$ The highest service delay margin
- 3: **return** $\text{fitness}(\pi) \leftarrow \phi_{max}$

π (line 5). FASD_MASC uses Algorithm 5 to calculate the fitness of a given VNF mapping and scheduling in a specific state. Fitness is determined based on the highest delay margin ϕ_{max} among active service requests, as indicated in line 2 of Algorithm 5. Consequently, the algorithm evaluates the fitness of both the current and next states (line 6). The next state replaces the current state if $\exp(\frac{\text{fitness}(\pi) - \text{fitness}(\pi^{Next})}{\mathcal{B}T}) > p$, where $\mathcal{B}$ represents the Boltzmann constant, and this value is compared against the predetermined acceptance probability threshold p . Upon completing L iterations, the algorithm decreases the current temperature by the cooling factor r (line 11).

Algorithm 4 outlines the procedure for generating the next state π^{Next} from the current state π . The algorithm starts by selecting a service request s_i from the set of active MC service requests $S(t)$ in the current state π (line 1). Next, it randomly selects a VNF component $f_{i,j} \in F_i$ of the chosen service s_i (line 2). Then, it identifies the source node v_k to which $f_{i,j}$ is currently mapped (with $x_k^{i,j} = 1$) in the current state π (line 3). Furthermore, it identifies potential target nodes $\mathcal{N}(f_{i,j})$ for relocation, considering resource availability and constraints outlined in Constraints (13f), (13g), and (13h) (line 4). Then, it randomly selects one of the candidate target nodes $v_h \in \mathcal{N}(f_{i,j})$ and performs the relocation or swapping operation by setting $x_k^{i,j} = 0$ and $x_h^{i,j} = 1$ (line 6). The algorithm updates the VNF node mapping strategy, X^{Next} , and then adjusts the virtual link mapping, Y^{Next} , by applying the Dijkstra shortest path selection algorithm from X^{Next} to account for the swap (line 8). Similarly, the VNF scheduling, Z^{Next} , is adjusted by applying the demand-based scheduling scheme from X^{Next} and Y^{Next} to account for the relocation (line 9). The final next state is generated by updating the next state with X^{Next} , Y^{Next} , and Z^{Next} (line 10).

V. PERFORMANCE EVALUATION

In this section, we compare our proposed algorithms with the benchmark algorithm TS_MAPSCH [34] and

TABLE III
SIMULATION PARAMETERS

Parameters	Value
Topology	SES O3b MEO constellation [18]
Number of SEC (K)	20 [18]
Satellite Ground Station Locations	9 [18]
Number of snapshots (M)	24 [18]
Service Arrival Rate(λ)	[0.05 – 0.20] requests/ms [34]
Processing delay of SEC node	[5 – 10] ms [34]
ISL propagation delay	[20 – 40] ms [18]
Uplink/downlink propagation delay	[29 – 75] ms [35]
Maximum tolerable E2E service delay	{200, 250, 300} ms [27], [28]
Number of VNFs per an SFC request	[3 – 5] [18]
Number of VNFs per an SEC node	[1 – 5] [18]
Processing capacity of an SEC node	96 vCPUs [22]
Storage capacity of an SEC node	112 GB [22]
Processing units required by a VNF	[2 – 4] vCPU [22]
Storage units required by a VNF	[4 – 8] GB [22]
Capacity of physical ISL	1 Gbps [22]
Capacity of physical uplink/downlink	1 Gbps [22]
Bandwidth demand of virtual link	[10 – 20] Mbps [22]

the optimization-based optimal INLP solution obtained using decomposition and relaxation methods discussed in Section III-D. The benchmark algorithm completes VNF mapping and scheduling of service requests on a *service-by-service* basis, indicating that the mapping and scheduling of one service must be finished before proceeding to the next one. In TS_MAPSCH, VNF re-mapping and re-scheduling occur only when there are violations in the E2E service delay demand.

A. Simulation Settings

Our simulations utilize the Medium Earth Orbit (MEO) constellation as the network topology, consisting of 20 SEC nodes and 9 satellite ground station locations, where users can submit service requests. We use the System Toolkit (STK) [35] to generate network topologies and data in each snapshot. As a result, we consider 24 snapshots (*i.e.*, $M = 24$) for the periodic satellite network with a recurrence period of 6 hours. Hence, each snapshot has a duration of 15 minutes. Furthermore, we assume 10 MC service types with specified VNF components, E2E delay, and bandwidth requirements. We assume that each MC service request is provisioned on the SEC nodes, characterized by a service type, a continuity indicator indicating whether it spans multiple snapshots and source-destination addresses. To ensure both heterogeneity and dynamism in traffic demand, we randomly generate service requests with varied parameters. We modeled the inter-arrival time among service requests using the Poisson distribution (*i.e.*, $M/M/1$ queuing model), expressed as: $t_I = -\frac{\log(1-R)}{\lambda}$, where R and λ represent the probability of a service request arrival and the mean arrival rate, respectively [36]. We choose a service arrival rate [0.05 – 0.20] requests/ms for mathematical simplicity [34], aligning with specified service delay demands in milliseconds [27], [28]. Additionally, the uplink and downlink propagation delays are randomly generated within the specified range outlined in Table III, showcasing how the delay varies based on the user's location on the ground. For FASD_MASC, SA parameters are set as: the initial temperature $T_0 = 100$, Markov chain length $L_0 = 200$, cooling rate $r = 0.97$, and acceptance probability $p = 0.85$ [37]. The remaining simulation parameters and their values are summarized in Table III.

TABLE IV
RUNNING TIME COMPARISON AT, $\lambda = 0.05$

Metrics	FAGD_MASC	FASD_MASC	TS_MAPSCH	INLP
Running Time	0.01357 <i>sec</i>	0.0876 <i>sec</i>	0.0872 <i>sec</i>	> 60 <i>sec</i>
$\phi_{max}(t) = \max_i \{\Phi(t)\}$	1.04791	1.04782	1.04791	1.04597

B. Running Time Comparison

In this subsection, we compare the proposed solutions with both the benchmark and optimization-based optimal INLP solutions in terms of their running time, considering the significance of this metric in network infrastructures with limited resources, such as SEC-enabled networks. Note that all algorithms run on DELL Precision 3551 laptop, Intel(R) Core(TM) RAM 32.0 GB i9-10885H CPU @ 2.40GHz.

As shown in Table IV, the proposed algorithms FAGD_MASC and FASD_MASC converge in approximately 0.01357 and 0.0876 seconds, respectively. In contrast, the optimization-based optimal INLP solution takes at least 60 seconds to converge. As a result, the optimization-based optimal solution for INLP is impractical to meet the demands of MC applications due to its significant computational time requirements, which do not align with the urgent and time-sensitive decision-making needs of MC applications. This motivated the proposal of FAGD_MASC and FASD_MASC algorithms to achieve reasonable results in shorter convergence times. Specifically, FAGD_MASC achieves faster convergence compared to the benchmark algorithm, while FASD_MASC shows a comparable convergence time.

C. Performance Difference Between the Proposed Solution and Optimization-Based Optimal INLP Solution

Initially, we employ the optimization solver CVX to solve the optimization-based optimal solution for INLP and obtain the optimal solution. For each snapshot ($G_1, G_2, \dots, G_{24}$), we conduct individual experiments, and the final results are obtained by calculating the average values across all snapshots. We evaluate the performance of the proposed algorithms in terms of their deviation from the optimization-based optimal INLP solution in three scenarios with varying arrival rates λ : *Low* ($\lambda = 0.05$), *Medium* ($\lambda = 0.1$), and *High* ($\lambda = 0.2$) traffic scales. We compare the proposed solutions against the optimization-based optimal INLP solution in terms of two metrics: service request acceptance rate and delay margin fairness.

Definition 2 (Service Acceptance Rate): We define this performance metric as the ratio of service requests that satisfy their E2E service delay requirement to the total number of service requests. We express a service request s_i acceptance function at snapshot t as $A_i(t)$ as:

$$A_i(t) = \begin{cases} 1, & \text{if } \phi_i(t) \leq 1, \\ 0, & \text{Otherwise.} \end{cases} \quad (18)$$

where $\phi_i(t)$ is the delay margin of the service request s_i . Therefore, the acceptance rate of the service requests at

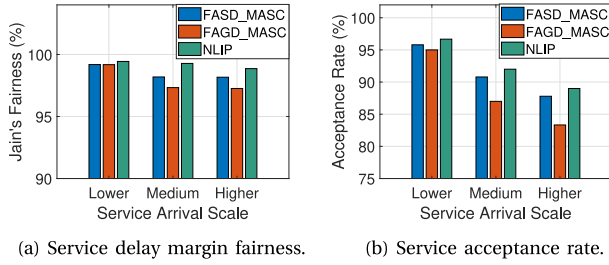


Fig. 2. Performance comparison of the proposed solutions with the optimization-based optimal INLP solution.

snapshot time t is expressed as [18]:

$$AR(t) = \frac{\sum_{i=1}^{|S(t)|} A_i(t)}{|S(t)|}, \quad (19)$$

Definition 3: The delay margin fairness is defined using Jain's fairness index $J(\phi(t))$ [38]:

$$J(\phi(t)) = \frac{\left(\sum_{i=1}^{|S(t)|} \phi_i(t) \right)^2}{|S(t)| \sum_{i=1}^{|S(t)|} \phi_i^2(t)}, \quad (20)$$

where $J(\phi(t))$ is the delay margin for $S(t)$ at t .

In Figure 2, the proposed algorithms exhibit a performance comparable to the optimization-based optimal INLP solution under low traffic arrival scale, with deviations of less than 0.70% and 0.07% in service acceptance rate and delay margin fairness, respectively. However, unlike the FAGD_MASC algorithm, FASD_MASC demonstrates robustness even with high traffic rates, showing less than 1.1% and 0.6% deviation in service acceptance rate and delay margin fairness compared to the optimization-based optimal INLP solution as shown in Figures (2b) and (2b), respectively.

D. Performance Comparison Between the Proposed Solutions and the Benchmark

In this subsection, we evaluate the performance of the proposed algorithms against the benchmark solution in terms of service request delay margin fairness, acceptance rate, and node resource utilization. We assess the results individually in each snapshot, and the final result is the average value in all snapshots.

1) Fairness for Varied Service Request Arrival Rates:

This study aims to maximize fairness in service delay margins between competing MC service requests. As shown in Figure 3(a), both the proposed algorithms FAGD_MASC and FASD_MASC outperform the benchmark by more than 14.6% and 19.5% in fairness for the arrival rates of $\lambda = 0.05$ and $\lambda = 0.2$, respectively. The reason for this is that the TS_MAPSCH algorithm performs VNF mapping and scheduling to achieve maximum profit for a service provider. It is obvious that maximum profit could be achieved while some requests are discriminated against, resulting in poor delay margin fairness. The proposed algorithms prioritize maintaining fairness in the delay margins for all requests, resulting in better fairness performance compared to the benchmark algorithm,

as evident in Figure 3(a). However, fairness is expected to decrease with increasing arrival rates. This is due to the challenge of finding link and node mapping strategies to maintain fairness when faced with an influx of MC service requests and limited network resources. However, the proposed algorithms consistently strive to maintain delay margin fairness, unlike the benchmark solution, resulting in gradual and slight degradation.

2) Acceptance Rates for Different Service Request Arrival Rates:

We compare the performance of the proposed algorithms with the benchmark in terms of the acceptance rate of service requests at different arrival rates. As depicted in Figure 3(b), the proposed solutions exhibit higher service acceptance rates compared to the benchmark. At a service rate of $\lambda = 0.05$, there is a gap of approximately 2.0% between the proposed algorithms and the benchmark. This gap widens further at higher service rates (e.g., at $\lambda = 0.2$, the gap exceeds 12.6%). This improvement comes from two main factors: i) Unlike TS_MAPSCH, FAGD_MASC and FASD_MASC employ a fine-grained VNF mapping and scheduling strategy at the VNF level, enhancing resource utilization efficiency to accommodate more service requests. ii) The topology-aware VNF mapping and scheduling strategies of the proposed algorithms result in accepting more service requests, considering factors such as the continuity indicator and link availability (as described in Equation (13l) and elaborated in Section (V-D4)).

3) Average Number of Nodes Used per Service: The number of nodes used per service is defined as the total number of nodes required to process the VNF components to complete a corresponding service request. As shown in Figure 3(c), the proposed algorithms accept service requests with fewer nodes than TS_MAPSCH. Unlike the benchmark, the proposed algorithms apply the VNF mapping and scheduling strategy at the VNF level instead of the service level (*service-by-service* scheme), which provides a more fine-grained strategy to deploy service requests with fewer SEC nodes and network links. For example, as shown in Figure 3(c), the number of nodes used to deliver service requests shows that the gap between the proposed algorithms and the benchmark reaches approximately 8.75% and 22.46% for $\lambda = 0.05$ and $\lambda = 0.20$, respectively. In other words, the FASD_MASC algorithm enables the provisioning of 100 service requests with at least 8-22 fewer nodes than the benchmark. Therefore, the proposed algorithms ensure better resource utilization than the benchmark, which consumes fewer nodes and links.

4) Acceptance Rate With Varying Service Request Continuity:

To assess the effect of dynamic satellite network topology on the acceptance rate of service requests spanning multiple snapshots, we compare the proposed algorithms with the benchmark. We initially analyze scenarios where no requests continue between snapshots (0% continuation rate), gradually increasing the continuation rate to 100% in each experiment. As illustrated in Figure 4, the service acceptance rate of the proposed algorithms experiences a slight decline with an increasing number of service requests continuing into the next snapshot. In contrast, the performance of the benchmark deteriorates notably as the proportion of continuing requests increases. For example, when no requests

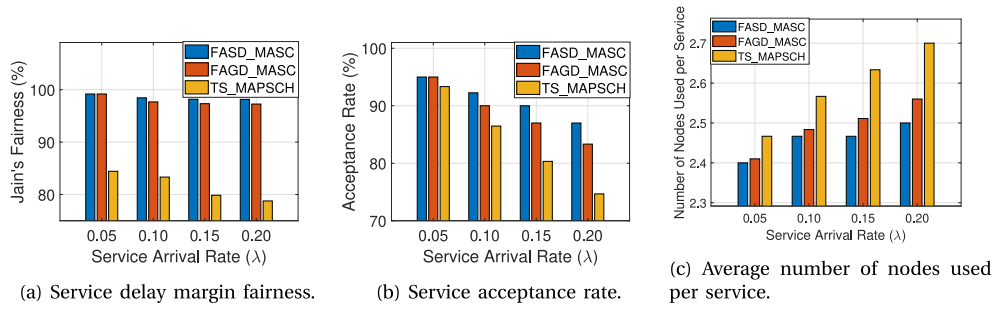


Fig. 3. Performance evaluation with varying service arrival rate.

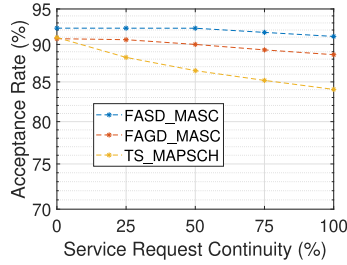


Fig. 4. Service acceptance rate continuity over snapshots.

continue to the next snapshot (*i.e.*, 0% of available requests), the benchmark achieves an acceptance rate of around 90.9%. However, when all service requests continue to the next snapshot (*i.e.*, 100% of available requests), the service request acceptance rate of the benchmark drops to approximately 83%, signifying a degradation of nearly 7.9% due to the time-varying topology. This degradation is expected to be more pronounced in highly dynamic network topologies like Low Earth Orbit (LEO) constellations. In contrast, the proposed algorithms show only a slight degradation of less than 0.81% because they consider both the state and availability of the link in their VNF mapping and scheduling strategies. Therefore, substantial improvements in the performance of the proposed algorithms over the benchmark can be attributed to their effective mitigation of time-varying topology challenges.

E. Algorithm Complexity

1) *Time Complexity Analysis for Algorithm 1*: In this algorithm, the innermost function, the *Dijkstra shortest path algorithm*, is invoked $\mathcal{O}(|F(t)|)$ times, where $|F(t)|$ is the maximum number of VNF components per service request. The time complexity of the *Dijkstra shortest path algorithm* is $\mathcal{O}(|E| + |V| \log |V|)$ in a graph $G = (V, E)$ [39], where $|V|$ and $|E|$ are the number of nodes and links in the graph topology, respectively. Therefore, Algorithm 1 has a time complexity of $\mathcal{O}(|F(t)| \cdot (|E| + |V| \log |V|))$, depending on the number of VNF components in a service request and the size of the network.

2) *Time Complexity Analysis for FAGD_MASC Algorithm*: If we denote I_{tr} as the maximum number of iterations to find the optimum solution, the *Dijkstra shortest path algorithm* is invoked I_{tr} times for each service request. Therefore, the time

TABLE V
ALGORITHM COMPLEXITY COMPARISON

Algorithm	Complexity
FAGD_MASC	$\mathcal{O}(I_{tr} \cdot S(t) \cdot F(t) \cdot (E + V \log V))$
FASD_MASC	$\mathcal{O}(L \log \frac{1}{r} T_0 \cdot S(t) \cdot F(t) \cdot (E + V \log V))$
TS_MAPSCH	$\mathcal{O}(I_{tr} \cdot M \cdot V \log V + S(t) \log S(t))$ [34]

complexity of Algorithm 2 can be defined as $\mathcal{O}(I_{tr} \cdot |S(t)| \cdot |F(t)| \cdot (|E| + |V| \log |V|))$, which mainly depends on the size of the network ($|V|$ and $|E|$), the number of requests ($|S(t)|$), and the number of VNF components per service request ($|F(t)|$).

3) *Time Complexity Analysis for FASD_MASC Algorithm*: The time complexity of the FASD_MASC algorithm depends on the SA parameters, including the initial temperature T_0 , cooling rate r , and Markov chain length L . The time complexity of an SA algorithm can be defined by $L \log \frac{1}{r} T_0$ [33]. Therefore, the time complexity of FASD_MASC shown in Algorithm 3 is $\mathcal{O}(L \log \frac{1}{r} T_0 \cdot |S(t)| \cdot |F(t)| \cdot (|E| + |V| \log |V|))$, which depends not only on the initial SA parameters but also on the sizes of the service requests and the network.

The TS_MAPSCH algorithm has a time complexity of $\mathcal{O}(|S(t)| \log |S(t)|)$ for sorting service sets and $\mathcal{O}(I_{tr} \cdot \omega \cdot |V| \log |V| + |S(t)| \log |S(t)|)$ overall, where ω is the number of neighbors and I_{tr} is the maximum allowed iterations.

The computational complexity of TS_MAPSCH is higher than that of FASD_MASC in terms of service requests $\mathcal{O}(|S(t)| \log |S(t)|)$. However, FAGD_MASC shares the same complexity as TS_MAPSCH in this aspect. Moreover, both the proposed algorithms and the benchmark have similar complexities regarding network size, such as the number of SEC nodes. While FASD_MASC exhibits higher computational complexity than FAGD_MASC due to its initial parameters, it performs steadily in high service arrival scenarios. Thus, FASD_MASC is preferable for high service arrival rates, while FAGD_MASC is better suited for low arrival rates due to its lower convergence time and comparable performance.

In summary, the proposed FASD_MASC outperforms benchmarks in service acceptance rate and fairness, with improvements of 10% and 15%, respectively. This superior performance is due to two main factors: (1) It facilitates VNF re-mapping and re-scheduling, dynamically adjusting strategies to match traffic demand and infrastructure changes. (2)

Its topological awareness, considering link state and availability, enhances performance. Besides its superior performance, the proposed solution's lower time complexity renders it well-suited for prompt service provisioning, especially for MC services. From an end-user perspective, this improvement ensures consistent service delivery, reduced latency, and higher reliability. For service providers, optimized resource utilization and increased service acceptance rates enhance operational efficiency and support a larger user base without compromising quality.

VI. CONCLUSION

In this article, we study dynamic VNF mapping and scheduling strategies in SDN/NFV-enabled satellite edge networks for MC applications. We proposed two heuristic algorithms, a greedy-based and an SA-based approach, to address the VNF mapping and scheduling problem in a dynamic NGSO system. The algorithms facilitate VNF re-mapping and re-scheduling, dynamically adjusting strategies to match traffic demand and infrastructure changes. Their topological awareness, considering link state and availability, results in superior performance compared to existing solutions. In conclusion, our study presents a novel VNF mapping and scheduling strategy for MC applications, advancing satellite edge computing for timely and reliable communication. The approach holds potential for applications in disaster response, remote medical care, and other time-sensitive operations. In future work, we will delve into the impact of VNF migration costs in CDN-based networks and explore MP-MP slices. Additionally, we aim to incorporate deployment time into our analysis.

APPENDIX A E2E SERVICE DELAY MODEL

The E2E service delay is the cumulative time taken to complete a service request via the selected physical path. The E2E service delay $d_i(t)$ for a specific service request s_i at a given snapshot time t , is defined as follows:

$$d_i(t) = \sum_{j=1}^{|F_i|} \left(\sum_{k=1}^{|V|} \left(\sum_{h=1}^{|V|} Y_{h,k}^{i,j}(t) d_{h,k}(t) \right) + X_k^{i,j}(t) \ell_k^{i,j}(t) + \tau_{i,j}^k(t) \right) \quad \forall e_{h,k} \in E, \forall v_k \in V. \quad (21)$$

The E2E service delay, $d_i(t)$, consists of three components.

- 1) The propagation delay of a service request includes the cumulative sum of the propagation delays of the physical links along the selected path required to fulfill the service request. The first component of the service delay $d_i(t)$ for the service request s_i is the propagation delay $d_i^{prop}(t)$, defined as follows:

$$d_i^{prop}(t) = \sum_{j=1}^{|F_i|} \left(\sum_{k=1}^{|V|} \left(\sum_{h=1}^{|V|} Y_{h,k}^{i,j}(t) d_{h,k}(t) \right) \right) \quad (22)$$

where $d_{h,k}(t)$ represents the propagation delay between the physical nodes v_h and v_k , and $Y_{h,k}^{i,j}(t)$ is the virtual link decision variable indicating whether the virtual link between VNF $f_{i,j}$ and its preceding VNF $f_{i,j-1}$ is mapped onto the path containing the link $e_{h,k}$ at the given snapshot t , respectively.

- 2) The second component, termed processing delay $d_i^{proc}(t)$ within $d_i(t)$, signifies the cumulative processing delay of the VNF components of s_i at their designated nodes. It is defined as:

$$d_i^{proc}(t) = \sum_{j=1}^{|F_i|} X_k^{i,j}(t) \ell_k^{i,j}(t) \quad (23)$$

where $\ell_k^{i,j}(t)$ stands for the processing delay necessary to handle the VNF component $f_{i,j}$ at node v_k , and $X_k^{i,j}(t)$ signifies the VNF mapping decision determining whether VNF component $f_{i,j}$ is executed at node v_k during the specified snapshot t .

- 3) We can define the queuing delay of a VNF as the total delay experienced by the VNF while it awaits processing at a designated node. This delay arises when the designated node is occupied with processing another VNF. The third component of $d_i(t)$ is the total queuing delay for VNF $f_{i,j}$ at node v_k , denoted as $\tau_{i,j}^k(t)$, and can be calculated as follows:

$$\begin{aligned} \tau_{i,j}^k(t) &= X_k^{i,j}(t) \\ &\sum_{g=1}^{|S(t)|} \sum_{q=1}^{|F_g|} \left(\left(\ell_k^{i,j} - (A_{g,q}^k(t) - A_{i,j}^k(t)) \right) \theta_{ij,gq}^k Z_{ij,gq}^k(t) \right. \\ &\quad \left. + \left(\ell_k^{g,q} - (A_{i,j}^k(t) - A_{g,q}^k(t)) \right) \theta_{gq,ij}^k Z_{gq,ij}^k(t) \right) \\ &\quad \forall \{g \neq i \text{ OR } q \neq j, f_{i,j} \in F_i, f_{g,q} \in F_g, v_k \in V\} \quad (24) \end{aligned}$$

where $\theta_{ij,gq}^k$ is a binary variable that shows whether VNF $f_{i,j}$ arrives while VNF $f_{g,q}$ waits for processing or is in the middle of processing at node v_k and can be described as follows.

$$\theta_{ij,gq}^k = \begin{cases} 1, & \text{if } f_{i,j} \text{ arrived while } f_{g,q} \text{ at } v_k \\ 0, & \text{Otherwise.} \end{cases}$$

From Equation (21), the E2E delay of a request can be categorized into two main components: VNF mapping delay component and VNF scheduling delay component.

The delay associated with VNF mapping encompasses the processing delay and propagation delay components, and it can be defined as follows:

$$d_i^m(t) = \sum_{j=1}^{|F_i|} \left(\sum_{k=1}^{|V|} \left(\sum_{h=1}^{|V|} Y_{h,k}^{i,j}(t) d_{h,k}(t) \right) + X_k^{i,j}(t) \ell_k^{i,j}(t) \right) \quad (25)$$

The delay attributed to VNF scheduling encompasses the cumulative queuing delay incurred by the VNF components within a service request. This delay can be defined as:

$$d_i^s(t) = \sum_{j=1}^{|F_i|} \tau_{i,j}^k(t) \quad (26)$$

APPENDIX B BINARY RELAXATION

Note that we define $X = X_k^{i,j}(t)$, $Y = Y_{h,k}^{i,j}(t)$ and $Z = Z_{ij,qq}^k(t)$ to simplify the solution of the optimization problem. We define the VNF mapping, ψ_{xy} and VNF scheduling, ψ_{xy} penalized functions which include the penalty functions and their corresponding parameters as:

$$\begin{aligned} \psi_{xy} = & \mu_x \sum_{j=1}^{|F_i|} \sum_{k=1}^{|V|} (X - X^{\alpha-1}) \nabla \mathcal{P}(X) \\ & + \mu_y \sum_{j=1}^{|F_i|} \sum_{h=1}^{|V|} \sum_{k=1}^{|V|} (Y - Y^{\alpha-1}) \nabla \mathcal{P}(Y) \quad (27) \end{aligned}$$

$$\psi_z = \mu_z \sum_{i=1}^{|S(t)|} \sum_{j=1}^{|F_i|} \sum_{g=1}^{|S(t)|} \sum_{q=1}^{|F_g|} (Z - Z^{\alpha-1}) \nabla \mathcal{P}(Z) \quad (28)$$

where $\mathcal{P}(X)$, $\mathcal{P}(Y)$, and $\mathcal{P}(Z)$, are relaxation penalty functions associated with the decision variables for VNF node mapping (X), virtual link mapping (Y), and VNF scheduling (Z), respectively and defined as:

$$\mathcal{P}(X) = X^2 - X, \mathcal{P}(Y) = Y^2 - Y, \text{ and } \mathcal{P}(Z) = Z^2 - Z \quad (29)$$

The penalty functions are designed to encourage the relaxed problem to generate solution values within the range of 0 to 1. Additionally, we introduce corresponding penalty parameters, represented as μ_x , μ_y , and μ_z , specifically for the decision variables $\mathbf{X}(t)$, $\mathbf{Y}(t)$, and $\mathbf{Z}(t)$, respectively. These penalty parameters play a role in shaping the behavior of the relaxation to encourage binary-like solutions. We assume that $X^{\alpha-1}$ and $Y^{\alpha-1}$ are the $(\alpha-1)^{th}$ VNF node and virtual link mapping solutions, respectively. Similarly, $Z^{\alpha-1}$ is considered the $(\alpha-1)^{th}$ VNF scheduling solution.

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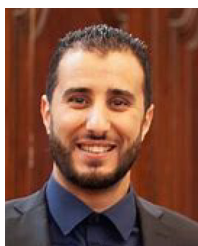
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Haftay Gebreslasie Abreha (Graduate Student Member, IEEE) received the M.Sc. degree in telecommunications engineering and communication network engineering jointly from the University of Trento and Scuola Superiore Sant'Anna, Italy, in 2015. He is currently a Doctoral Researcher with the Interdisciplinary Centre for Security, Reliability and Trust (SnT), University of Luxembourg. Before joining SnT, he was a Research Assistant at renowned institutions, including IMDEA Networks Institute, Spain, the University of Trento, Italy, and the United

Arab Emirates University. He also has industrial experience as a network administrator with the Immigration and Citizenship Service of Ethiopia. His research focuses on edge computing architectures for integrated satellite-terrestrial networks, with a particular emphasis on optimizing traffic engineering algorithms for software-defined networking, and network function virtualization for efficient service provisioning.



Houcine Chougrani (Member, IEEE) received the State Engineering degree in electronics from École Nationale Polytechnique Algiers, Algeria, in 2011, the M.S. degree in signal processing from Paul Sabatier University, Toulouse, France, in 2012, and the Ph.D. degree in information technology from École Telecom Bretagne, Brest, France, in 2016. He is a Research Scientist with the SnT, University of Luxembourg. From 2017 to 2018, he was a Research and Development Engineer with the Department of Telecommunication, Airbus Defence and Space.

From 2018 to 2019, he was a Research and Development Engineer with OQ Technology. His research interests include the development and implementation of advanced techniques for both terrestrial and satellite communication focusing on the 3GPP-based protocols.



Ilora Maity (Member, IEEE) received the B.Tech. degree in computer science and engineering from the West Bengal University of Technology, India, in 2008, the M.E. degree in computer science and engineering from the Indian Institute of Engineering Science and Technology, Shibpur, India, in 2011, and the Ph.D. degree from the Indian Institute of Technology Kharagpur, India, in 2021. She has worked as a Technical Analyst of Banking & Financial Services with Cognizant Technology Solutions India Pvt. Ltd., Kolkata, India, from September 2011 to July 2015. From January 2021 to April 2021, she was a Postdoctoral Researcher with Aalto University, Finland. She is a Research Associate with the Interdisciplinary Centre for Security, Reliability and Trust, University of Luxembourg. She has authored over 30 technical papers in refereed international journals and conferences. Her research interests include software-defined networking, network function virtualization, the Internet of Things, satellite communications, and quantum communications. She also served as a TPC Member for several IEEE conferences. She is an Editorial Board of *SciencePG International Journal of Wireless Communications and Mobile Computing*.



Youssef Drif (Member, IEEE) received the Ph.D. degree from the Federal University of Toulouse Midi-Pyrénées, France, in 2022. He joined the Signal Processing & Satellite Communications Research Group, SIGCOM, headed by Prof. Chatzinotas, and has since worked on 5G-NTN, UAVs, and Quantum technologies in multiple ESA and European projects. His research interests are SDN and NFV for Satcom and, more precisely, in using these technologies for satellite integration into mobile networks.



Christos Politis (Fellow, IEEE) received the Dipl.-Eng. degree in electrical and computer engineering from the University of Patras, Greece, in 2011, the first M.Sc. degree in space communication systems from ISAE-SUPAERO, France, in 2012, the second M.Sc. degree in communications and signal processing from the University of Newcastle, U.K. in 2014, and the Ph.D. degree in spectrum monitoring algorithms for wireless and satellite communications from the University of Luxembourg, Luxembourg, in 2018. He is a Senior Communications Systems

Engineer with SES TechCom. Prior to joining SES in January 2018, he was a Ph.D. Researcher with the Interdisciplinary Centre for Security, Reliability and Trust, University of Luxembourg from 2014 to 2017, and a Satellite Communications Engineer with Centre National d'Etudes Spatiales, France, in 2012. He has technical experience in national and international Research and Development and Innovation projects on ICT with major focus on Satellite Communications and he has authored several journals, conference and book chapter publications. In particular, he has been contributing to the 5G related projects H2020 Work Program from 2018 to 2020 "INGENIOUS", H2020 5G PPP Phase III "5G-VINNI", H2020 5G PPP Phase II "SaT5G", ESA ARTES "SATIS5" and ESA ARTES "EdgeSAT", as well as to the phase 1 of ESA funded "5G-EMERGE".



Symeon Chatzinotas (Fellow, IEEE) is a Full Professor and the Head of the SIGCOM Research Group, SnT, University of Luxembourg. In the past, he has been a Visiting Professor with the University of Parma, Italy, and was involved in numerous RD projects for NCSR Demokritos, CERTH Hellas, and CCSR, University of Surrey. He has co-authored more than 700 technical papers in refereed international journals, conferences and scientific books. He is coordinating the research activities on communications and networking across a group of

80 researchers, acting as a PI for more than 40 projects and main representative for 3GPP, ETSI, and DVB. He was the co-recipient of the 2014 IEEE Distinguished Contributions to Satellite Communications Award and the Best Paper Awards at WCNC, 5GWF, EURASIP JWCN, CROWNCOM, and ICSSC. He is currently serving an Editorial Board for the IEEE TRANSACTIONS ON COMMUNICATIONS, IEEE OPEN JOURNAL OF VEHICULAR TECHNOLOGY, and *International Journal of Satellite Communications and Networking*.